

ASSIGNMENT #06

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Name : Liv Lyons

Due Date: Friday, April 17th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Thermodynamics Report 6

Liv Lyons

Dr. Caracotsios

April 17, 2026

Contents

Abstract	2
Background	2
Given Information	3
Problem A	4
0.1 Determination of Water Content, Dew Point, and Bubble Point Using Peng-Robinson EOS	4
0.1.1 Calculation for Water Content	7
0.1.2 Dew Point Temperature Calculation	11
0.1.3 Bubble Point Temperature Calculation	11
Problem B	13
0.2 Fugacity Analysis of a Ternary Mixture Using Ideal and Peng-Robinson Approaches	13
0.2.1 Verifying Liquid Phase – Ideal	14
0.2.2 Fugacity Calculations – Ideal	14
0.2.3 Verifying Liquid Phase – Non-Ideal	15
0.2.4 Fugacity Calculations – Non-Ideal	16
Conclusion	18
References	20

Abstract

In this work, vapor–liquid equilibrium (VLE) calculations were performed for two multi-component systems using the Peng–Robinson equation of state (PR EOS). In Problem A, a natural gas mixture was analyzed to determine the maximum amount of water that can be present in the vapor phase at specified conditions, expressed in lb/MMSCF. This was accomplished by applying phase equilibrium criteria and solving for the water mole fraction at saturation. The corresponding dew point and bubble point temperatures were also determined, representing the onset of liquid formation and vapor formation, respectively.

In Problem B, fugacities of components in a ternary liquid mixture of methanol, ethanol, and n-butanol were evaluated under both ideal and non-ideal assumptions. The ideal solution behavior was compared with predictions from the Peng–Robinson EOS, allowing for assessment of non-ideal effects. Additionally, phase behavior was verified by calculating bubble point and dew point temperatures.

These calculations demonstrate the application of thermodynamic principles and equations of state to predict phase behavior and quantify deviations from ideality in multicomponent systems.

Background

Vapor–liquid equilibrium (VLE) describes the condition under which liquid and vapor phases coexist in thermodynamic equilibrium. At equilibrium, the fugacity of each component in the vapor phase is equal to that in the liquid phase:

$$f_i^V = f_i^L \quad (1)$$

For real systems, fugacity is expressed using fugacity coefficients:

$$f_i = y_i P \phi_i^V \quad (\text{vapor phase}), \quad f_i = x_i P \phi_i^L \quad (\text{liquid phase}) \quad (2)$$

In this work, the Peng–Robinson equation of state is used to calculate compressibility factors and fugacity coefficients for both phases. This equation of state accounts for non-ideal behavior in high-pressure systems such as natural gas mixtures.

Given Information

Problem A: Natural Gas Mixture

The natural gas composition is provided in weight percent and converted to mole fractions for analysis:

$$z_{CH_4} = 0.800$$

$$z_{C_2H_6} = 0.100$$

$$z_{C_3H_8} = 0.040$$

$$z_{iC_4H_{10}} = 0.030$$

$$z_{nC_4H_{10}} = 0.030$$

The system conditions are:

- Pressure: $P = 900$ psia or $P = 62.0526$ bar
- Temperature: $T = 65^\circ\text{F}$ or $T = 18.3333^\circ\text{C}$

The objective is to determine the maximum water content in the gas phase (lb/MMSCF) and the corresponding dew point and bubble point temperatures. The liquid phase is assumed to consist of pure water, and all calculations are performed using the Peng–Robinson EOS with zero binary interaction parameters.

Problem B: Ternary Liquid Mixture

A ternary mixture consisting of methanol, ethanol, and n-butanol is given with the following composition:

$$x = [0.2, 0.3, 0.5] \tag{3}$$

The system conditions are:

- Temperature: $T = 25^\circ\text{C}$

- Pressure: $P = 20$ bar

The objectives are to verify the phase of the system by calculating bubble and dew point temperatures and to determine fugacities under both ideal and non-ideal assumptions using the Peng–Robinson equation of state.

Problem A

0.1 Determination of Water Content, Dew Point, and Bubble Point Using Peng-Robinson EOS

Problem Overview

A natural gas mixture of known composition is analyzed at a specified pressure and temperature. The objectives are:

1. Determine the maximum amount of water the gas can hold (lb/MMSCF)
2. Determine the dew point temperature
3. Dertermine the bubble point temperature

The system is modeled using the Peng–Robinson Equation of State (PR EOS) with zero binary interaction parameters.

Phase Equilibrium Condition

At vapor–liquid equilibrium (VLE), the fugacity of each component must be equal in all phases:

$$f_i^V = f_i^L \quad (4)$$

In this system:

- The vapor phase contains hydrocarbons and water

- The liquid phase consists of pure water only

Thus, the equilibrium condition reduces to water:

$$f_w^V = f_w^L \quad (5)$$

Fugacity Expressions

Using fugacity coefficients:

Vapor phase:

$$f_w^V = y_w, P, \phi_w^V \quad (6)$$

Liquid phase (pure water):

$$f_w^L = P, \phi_w^L \quad (7)$$

Setting these equal:

$$y_w, P, \phi_w^V = P, \phi_w^L \quad (8)$$

Canceling pressure:

$$y_w, \phi_w^V = \phi_w^L \quad (9)$$

Solving for the vapor-phase mole fraction of water:

$$y_w = \frac{\phi_w^L}{\phi_w^V} \quad (10)$$

Derivation of Vapor-Phase Composition Adjustment

The given composition corresponds to a dry gas mixture:

$$\sum z_i = 1 \quad (11)$$

When water is introduced into the vapor phase, the total mole fraction must satisfy:

$$\sum y_i + y_w = 1 \quad (12)$$

Rearranging:

$$\sum y_i = 1 - y_w \quad (13)$$

Assuming that the relative proportions of hydrocarbons remain unchanged upon addition of water, the vapor-phase composition is proportional to the original dry composition:

$$y_i \propto z_i \quad (14)$$

Thus, we write:

$$y_i = C, z_i \quad (15)$$

Summing over all hydrocarbons:

$$\sum y_i = C \sum z_i = C \quad (16)$$

Since $\sum z_i = 1$, we obtain:

$$C = 1 - y_w \quad (17)$$

Therefore, the vapor-phase mole fractions are:

$$y_i = z_i(1 - y_w) \quad (18)$$

Physical Interpretation:

This expression indicates that the water mole fraction y_w occupies part of the total mixture, and the remaining fraction $(1 - y_w)$ is distributed among the hydrocarbons in proportion to their original dry composition.

0.1.1 Calculation for Water Content

Because the fugacity coefficient in the vapor phase depends on composition, which itself depends on y_w , the solution requires iteration.

1. Assume an initial value for y_w
2. Update hydrocarbon compositions:

$$y_i = z_i(1 - y_w)$$

3. Use the Peng–Robinson EOS to compute:

- Vapor compressibility factor Z^V
- Vapor fugacity coefficient ϕ_w^V

4. Compute liquid phase properties for pure water:

- Liquid compressibility factor Z^L
- Liquid fugacity coefficient ϕ_w^L

5. Update the water mole fraction:

$$y_w^{\text{new}} = \frac{\phi_w^L}{\phi_w^V}$$

6. Repeat until convergence:

$$|y_w^{\text{new}} - y_w^{\text{old}}| \rightarrow 0$$

Sample Calculation for Water Mole Fraction

From the phase equilibrium condition:

$$y_w = \frac{\phi_w^L}{\phi_w^V} \tag{19}$$

Using values obtained from the Peng–Robinson EOS:

$$\phi_w^V = 0.3886$$

$$\phi_w^L = 0.0003$$

Substituting:

$$y_w = \frac{0.0003}{0.3886} = 7.72 \times 10^{-4} \quad (20)$$

$$y_w \approx 0.00077 \quad (21)$$

Verification of Fugacity Equality

To confirm equilibrium, the fugacity of water in both phases is evaluated.

Vapor phase:

$$f_w^V = y_w * P * \phi_w^V \quad (22)$$

Substituting values:

$$\begin{aligned} f_w^V &= (0.000766)(62.0526)(0.3886) \\ &= 0.0184 \text{ bar} \end{aligned}$$

Liquid phase:

$$f_w^L = P, \phi_w^L \quad (23)$$

$$\begin{aligned} f_w^L &= (62.0526)(0.0003) \\ &= 0.0186 \text{ bar} \end{aligned}$$

Discussion of Results

The fugacities of water in the vapor and liquid phases are nearly equal:

$$f_w^V \approx f_w^L \quad (24)$$

This confirms that the system is at equilibrium and that the calculated value of y_w corresponds to the saturation condition.

Updated Vapor Composition

Using the relation:

$$y_i = z_i(1 - y_w) \quad (25)$$

and $y_w = 0.000766$:

$$1 - y_w = 0.999234$$

$$y_{CH_4} = 0.800(0.999234) = 0.79939$$

$$y_{C_2H_6} = 0.100(0.999234) = 0.09992$$

$$y_{C_3H_8} = 0.040(0.999234) = 0.03997$$

$$y_{iC_4H_{10}} = 0.030(0.999234) = 0.02998$$

$$y_{nC_4H_{10}} = 0.030(0.999234) = 0.02998$$

These values match those obtained in the spreadsheet and satisfy:

$$\sum y_i + y_w = 1$$

	A	B	C	D	E	F	G	H	I
1	VLE Calculations		Peng-Robinson						
2	Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O	
3	Molar Composition	1.0010	0.800000	0.100000	0.040000	0.030000	0.030000	0.001000	
4	Temperature, C	18.3333							
5	Pressure, bar	62.0526							
6	NEW MOLAR COMP	1.0000	0.799388	0.099923	0.039969	0.029977	0.029977	0.000766	
7	Saturated Vapor Pressure, bar								
8	Bubble Point Pressure, bar								
9	Dew Point Pressure, bar								
10									
11	Saturated Vapor Temperature, C								
12	Bubble Point Temperature, C								
13	Dew Point Temperature, C	18.3333							
14									
15	K-Values								
16	New K-Values								
17									
18	Vapor to Feed Molar Ratio ($\alpha=V/F$)								
19	The Rachford-Rice Equation								
20	Vapor Phase	Mixture							
21	Molar Composition	1.0010	0.800000	0.100000	0.040000	0.030000	0.030000	0.001000	
22	New Molar Composition	1.0000	0.799387	0.099923	0.039969	0.029977	0.029977	0.000766	
23	Compressibility Factor	0.7492							
24	Fugacity Coefficient		0.8873	0.5867	0.4166	0.2969	0.3174	0.3886	
25	Saturated Fugacity Coefficient								
26									
27	Molar Enthalpy, kJ/kmol								
28	Molar Entropy, kJ/(kmol K)								
29	Molar Density, kmol/m³								
30	Molar Heat Capacity, kJ/(kmol K)								
31	Liquid Phase	Mixture							
32	Molar Composition	1.0000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	
33	New Molar Composition	1.0000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	
34	Compressibility Factor	0.0541							
35	Fugacity Coefficient							0.0003	
36	Liquid Activity Coefficient								
37									
38	Molar Enthalpy, kJ/kmol								
39	Molar Entropy, kJ/(kmol K)								
40	Molar Density, kmol/m³								
41	Molar Heat Capacity, kJ/(kmol K)								
42	Molar Heat of Vaporization, kJ/kmol								
43	User Data and Calculations: ON								
44	yw	0.000765							
45	1-yw	0.9992							
46									

Figure 1: Water Molar Composition Calculations

The value we obtain is 0.000766 kmol water/kmol gas for the molar composition of water.

Conversion to lb/MMSCF

The calculated mole fraction of water can be interpreted as kmol of water per kmol of gas. To express this as lb water/MMSCF, the amount of water is first converted to mass by multiplying by the molecular weight of water and the factor 2.204623 to convert kilograms to pounds. The corresponding gas volume is obtained by converting kmol of gas to MMSCF using the CHE calculator function =KMOLtoMMSCF. Using this approach, the water content is determined to be:

$$36.3221 \text{ lb water/MMSCF}$$

(26)

0.1.2 Dew Point Temperature Calculation

At the dew point:

- Vapor composition is fixed
- The first drop of liquid forms

The equilibrium condition becomes:

$$y_w, \phi_w^V(T) = \phi_w^L(T) \quad (27)$$

This condition represents the point at which the first infinitesimal amount of liquid water forms from the vapor phase. Therefore, the associated temperature corresponds to the dew-point temperature:

$$\boxed{T_{\text{dew}} = 18.33^\circ\text{C}} \quad (28)$$

0.1.3 Bubble Point Temperature Calculation

The bubble point temperature corresponds to the condition at which the first infinitesimal amount of vapor forms from a liquid mixture. At this point, the liquid phase composition is known, and the vapor phase begins to form in equilibrium with it.

The phase equilibrium condition remains:

$$f_i^L = f_i^V \quad (29)$$

Using fugacity coefficients, this becomes:

Liquid phase:

$$f_i^L = x_i, P, \phi_i^L \quad (30)$$

Vapor phase:

$$f_i^V = y_i, P, \phi_i^V \quad (31)$$

Setting these equal:

$$x_i, \phi_i^L = y_i, \phi_i^V \quad (32)$$

Simplification for This System

In this problem, the liquid phase consists entirely of water:

$$x_w = 1, \quad x_i = 0; \text{ for hydrocarbons} \quad (33)$$

Thus, the equilibrium condition reduces to:

$$\phi_w^L = y_w, \phi_w^V \quad (34)$$

which is identical to the relation used in the dew point calculation.

Solution Procedure

To determine the bubble point temperature:

1. Assume an initial temperature
2. Compute the fugacity coefficient of water in the liquid phase, ϕ_w^L
3. Compute the fugacity coefficient of water in the vapor phase, ϕ_w^V
4. Evaluate the equilibrium condition:

$$F(T) = \phi_w^L - y_w \phi_w^V$$

5. Using the Bubble Point tool in the CHE calculator ribbon, the software automatically repeats this procedure.

Because the liquid phase contains only water, the bubble point temperature corresponds closely to the saturation temperature of pure water at the specified pressure. As a result, the calculated bubble point temperature is significantly lower than the dew point temperature of the gas mixture, reflecting the fact that hydrocarbons remain in the vapor phase while only water undergoes phase transition.

$$T_{\text{bubble}} = -57.34^{\circ}\text{C} \quad (35)$$

VLE Calculations		Peng-Robinson					
Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O
Molar Composition	1.0000	0.799388	0.099923	0.039969	0.029977	0.029977	0.000766
Temperature, C	18.3333						
Pressure, bar	62.0526						
Saturated Vapor Pressure, bar							
Bubble Point Pressure, bar							
Dew Point Pressure, bar							
Saturated Vapor Temperature, C							
Bubble Point Temperature, C	-57.3421						
Dew Point Temperature, C							
K-Values		1.1816	0.4165	0.1926	0.0910	0.1128	0.0734
New K-Values							
Vapor to Feed Molar Ratio ($\alpha=V/F$)	0.0000						
The Rachford-Rice Equation	0.0000						
Vapor Phase							
Molar Composition	1.0000	0.944525	0.041616	0.007698	0.002726	0.003380	0.000056
New Molar Composition		0.944524	0.041617	0.007698	0.002726	0.003380	0.000056
Compressibility Factor	0.4687						
Fugacity Coefficient		0.6827	0.2325	0.0954	0.0397	0.0474	0.0791
Saturated Fugacity Coefficient							
Molar Enthalpy, kJ/kmol							
Molar Entropy, kJ/(kmol K)							
Molar Density, kmol/m ³							
Molar Heat Capacity, kJ/(kmol K)							
Liquid Phase							
Molar Composition	1.0000	0.799388	0.099923	0.039969	0.029977	0.029977	0.000766
New Molar Composition							
Compressibility Factor	0.2117						
Fugacity Coefficient		0.8066	0.0968	0.0184	0.0036	0.0053	0.0058
Liquid Activity Coefficient							

Figure 2: Problem A Bubble Point Calculation

Problem B

0.2 Fugacity Analysis of a Ternary Mixture Using Ideal and Peng-Robinson Approaches

We know we have a ternary mixture of methanol, ethanol, and *n*-butanol. Their molar compositions are 0.2, 0.3, and 0.5, respectively.

This problem involves comparing a simple thermodynamic model (ideal) with a more complex one (Peng–Robinson). Because the pressure ($P = 20$ bar) is much higher than the vapor pressures of these alcohols at 25°C , the Poynting factor is very important.

0.2.1 Verifying Liquid Phase – Ideal

To verify that the mixture is a liquid at 25°C and 20 bar, we need to show that the system temperature is below the bubble point temperature at that pressure.

Modifying Raoult's Law:

$$P = \sum_{i=1}^3 x_i P_i^{\text{sat}}(T) \cdot PF_i \quad (36)$$

where PF_i is the Poynting factor:

$$PF_i = \exp\left(\frac{V_i^L(P - P_i^{\text{sat}})}{RT}\right) \quad (37)$$

We need to iterate on T until the sum equals 20 bar.

We find that T_{bub} is significantly higher than 25°C, confirming the mixture is a subcooled liquid.

These values are for ideal cases:

$$\text{Dew Point Temperature} = 214.33^\circ\text{C}$$

$$\text{Bubble Point Temperature} = 198.95^\circ\text{C}$$

0.2.2 Fugacity Calculations – Ideal

For an ideal solution, the fugacity of a component in the liquid mixture is simply its mole fraction multiplied by the fugacity of the pure liquid at T and P .

Ideal Pure Component:

$$f_i = P_i^{\text{sat}} \cdot \phi_i^{\text{sat}} \cdot \exp\left(\frac{V_i^L(P - P_i^{\text{sat}})}{RT}\right) \quad (38)$$

Since P_i^{sat} is low, we assume:

$$\phi_i^{\text{sat}} \approx 1$$

Ideal Component in Mixture:

$$\hat{f}_i = x_i \cdot f_i \quad (39)$$

0.2.3 Verifying Liquid Phase – Non-Ideal

VLE Calculations		Peng-Robinson						
Fluid Conditions	Mixture	CH4O	C2H6O	C4H10O				
Molar Composition	1.0000	0.200000	0.300000	0.500000				
Temperature, C	25.0000							
Pressure, bar	20.0000							
Saturated Vapor Pressure, bar		0.1587	0.0729	0.0100				
Bubble Point Pressure, bar								
Dew Point Pressure, bar								
Saturated Vapor Temperature, C								
Bubble Point Temperature, C								
Dew Point Temperature, C	214.3310							
K-Values		1.9582	1.5630	0.7083				
New K-Values								
Vapor to Feed Molar Ratio ($\alpha=V/F$)	1.0000							
The Rachford-Rice Equation	0.0000							
Vapor Phase	Mixture							
Molar Composition	1.0000	0.200000	0.300000	0.500000				
New Molar Composition								
Compressibility Factor	0.7632							
Fugacity Coefficient		0.9114	0.8646	0.7360				
Saturated Fugacity Coefficient								
Molar Enthalpy, kJ/kmol								
Molar Entropy, kJ/(kmol K)								
Molar Density, kmol/m3								
Molar Heat Capacity, kJ/(kmol K)								
Liquid Phase	Mixture							
Molar Composition	1.0000	0.102132	0.191942	0.705926				
New Molar Composition	1.0000	0.102132	0.191942	0.705926				
Compressibility Factor	0.0592							
Fugacity Coefficient		1.7847	1.3514	0.5213				
Liquid Activity Coefficient								
Molar Enthalpy, kJ/kmol								

Figure 3: Peng-Robinson Dew Point Temperature Calculation

VLE Calculations				
Fluid Conditions	Mixture	CH ₄ O	C ₂ H ₆ O	C ₄ H ₁₀ O
Molar Composition	1.0000	0.200000	0.300000	0.500000
Temperature, °C	25.0000			
Pressure, bar	20.0000			
Saturated Vapor Pressure, bar		0.1587	0.0729	0.0100
Bubble Point Pressure, bar				
Dew Point Pressure, bar				
Saturated Vapor Temperature, °C				
Bubble Point Temperature, °C	198.9528			
Dew Point Temperature, °C				
K-Values		1.6635	1.2982	0.5557
New K-Values				
Vapor to Feed Molar Ratio [$\alpha=V/F$]	0.0000			
The Rachford-Rice Equation	0.0000			
Vapor Phase	Mixture			
Molar Composition	1.0000	0.332695	0.389450	0.277855
New Molar Composition	1.0000	0.332695	0.389450	0.277855
Compressibility Factor	0.7836			
Fugacity Coefficient		0.8872	0.8417	0.7188
Saturated Fugacity Coefficient				
Molar Enthalpy, kJ/kmol				
Molar Entropy, kJ/(kmol K)				
Molar Density, kmol/m ³				
Molar Heat Capacity, kJ/(kmol K)				
Liquid Phase	Mixture			
Molar Composition	1.0000	0.200000	0.300000	0.500000
New Molar Composition				
Compressibility Factor	0.0535			
Fugacity Coefficient		1.4759	1.0927	0.3994
Liquid Activity Coefficient				

Figure 4: Peng-Robinson Bubble Point Temperature Calculation

0.2.4 Fugacity Calculations – Non-Ideal

For the Peng–Robinson case, fugacity is calculated using the fugacity coefficient.

Pure Component:

We set one component's molar composition to 1 and the others to 0. We then repeat this for each component and multiply the resulting fugacity coefficient by pressure:

$$f_i = P \cdot \phi_i^{\text{pure}} \quad (40)$$

Using CH₄O as an example:

VLE Calculations				
Fluid Conditions	Mixture	CH4O	C2H6O	C4H10O
Molar Composition	1.0000	1.000000	0.000000	0.000000
Temperature, C	25.0000			
Pressure, bar	20.0000			
Saturated Vapor Pressure, bar		0.1587	0.0729	0.0100
Bubble Point Pressure, bar				
Dew Point Pressure, bar				
Saturated Vapor Temperature, C				
Bubble Point Temperature, C				
Dew Point Temperature, C	214.3310			
K-Values		0.0118	0.0054	0.0010
New K-Values				
Vapor to Feed Molar Ratio [$\alpha=V/F$]	1.0000			
The Rachford-Rice Equation	-83.8994			
Vapor Phase	Mixture			
Molar Composition	1.0000	1.000000	0.000000	0.000000
New Molar Composition				
Compressibility Factor	0.8798			
Fugacity Coefficient		0.8907	0.8608	0.7752
Saturated Fugacity Coefficient				
Molar Enthalpy, kJ/kmol				
Molar Entropy, kJ/(kmol K)				
Molar Density, kmol/m ³				
Molar Heat Capacity, kJ/(kmol K)				
Liquid Phase	Mixture			
Molar Composition	1.0000	0.102132	0.191942	0.705926
New Molar Composition	84.8994	84.899405	0.000000	0.000000
Compressibility Factor	0.0664			
Fugacity Coefficient		0.0105	0.0046	0.0008
Liquid Activity Coefficient				
Molar Enthalpy, kJ/kmol				
Molar Entropy, kJ/(kmol K)				
Molar Density, kmol/m ³				
Molar Heat Capacity, kJ/(kmol K)				
Molar Heat of Vaporization, kJ/kmol				
User Data and Calculations: ON				
Pure Component		0.2098	0.0923	0.0155

Figure 5: CH₄O Example for Pure Component Calculation

Note: We must reference 25°C as the temperature for fugacity calculations.

Component in Mixture:

We use the original compositions ($z = [0.2, , 0.3, , 0.5]$) in this case:

$$\hat{f}_i = x_i \cdot P \cdot \hat{\phi}_i^{\text{mix}} \quad (41)$$

Note: Since the alcohols are polar, the standard Peng–Robinson equation may show some deviation from the ideal model, even with zero binary interaction parameters.

	IDEAL SOLUTION		NON-IDEAL SOLUTION (PENG-ROBINSON)	
	Pure Component	Component in Mixture	Pure Component	Component in Mixture
Methanol	0.1639	0.0328	0.2098	0.0420
Ethanol	0.0764	0.0229	0.0923	0.0277
N-Butanol	0.0108	0.0054	0.0155	0.0077

Figure 6: Fugacity Calculations

Conclusion

In this work, vapor–liquid equilibrium behavior and fugacity relationships were analyzed for multicomponent systems using both ideal models and the Peng–Robinson equation of state (PR EOS).

For the natural gas system in Problem A, the maximum water content was determined by enforcing phase equilibrium between the vapor mixture and pure liquid water. This approach allowed for the calculation of the water mole fraction at saturation, which was then converted to engineering units of lb/MMSCF. The corresponding dew point temperature was identified as the condition at which the first drop of liquid water forms, while the bubble point temperature represented the onset of vapor formation from the liquid phase. The results highlight how even small amounts of water can significantly impact phase behavior at high pressures.

In Problem B, fugacity calculations were performed for a ternary mixture of methanol, ethanol, and *n*-butanol under both ideal and non-ideal assumptions. The ideal model, incorporating Raoult’s Law with the Poynting correction, provided a baseline for comparison, while the Peng–Robinson EOS accounted for non-ideal interactions. Phase verification confirmed that the system exists as a subcooled liquid at the specified conditions. Comparison between the two approaches demonstrated that non-ideal effects become important for polar components and at elevated pressures, leading to deviations from ideal behavior.

Overall, this study illustrates the importance of applying appropriate thermodynamic models when analyzing real systems. While ideal assumptions can provide useful initial estimates, equations of state such as Peng–Robinson are necessary for accurately predicting phase behavior and fugacity in non-ideal, high-pressure mixtures.

References

1. Caracotsios, M. *An Introduction to Chemical Engineering Thermodynamics: Theory and Applications*. First Edition. 2026.
2. OpenAI. *ChatGPT*. Assistance with thermodynamic derivations, equation formatting, and LaTeX implementation. 2026.
3. Google Search. Used for verification of literature values for nitrogen thermodynamic properties and speed of sound at 10°C . 2026.